FORMAL COMPLETION ALONG A CLOSED SUBSET PRESERVES COUNTABLE INDEXEDNESS, THE ADIC* PROPERTY AND NOETHERIANNESS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The last sentence of the proof is garbled in the author's own text: it reads 'if X is locally countably indexed, or locally adic*, pr locally Noetherian, the is X_i is countably index, or adic*, or Noetherian'. It is reproduced verbatim, as the delivered text is the author's; no rewording or correction was inserted.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The definition of a formal algebraic space through a presentation, and its locality. Same chapter, Definition definition-formal-algebraic-space, invoked at proof lines 8166-8168 and 8170-8172. Supplies the covering along which the statement is checked.
- The base change of a completion is the completion along the inverse image. Same chapter, Lemma lemma-map-recompletions, invoked at proof lines 8168-8170. Identifies the base changes of the completion as completions of the members.
- The affine case: completion along a closed subset preserves the three types. Same chapter, Lemma lemma-recompletion-affine-types, invoked at proof lines 8177-8179. Gives the conclusion once the hypotheses have been transferred.

Source reference. Original lemma 0GVT: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 8046–8046. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GVV: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 8094–8094. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i \in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let X be a formal algebraic space over S. Let $T \subset X_{red}$ be a closed subset and let $X_{/T}$ be the formal completion of X along T. Then*

(1) *if $X_{red} \setminus T \to X_{red}$ is quasi-compact and X is locally countably indexed, then $X_{/T}$ is locally countably indexed,*
(2) *if $X_{red} \setminus T \to X_{red}$ is quasi-compact and X is locally adic*, then $X_{/T}$ is locally adic*, and*
(3) *if X is locally Noetherian, then $X_{/T}$ is locally Noetherian.*

Proof. Choose a covering $\{X_i \to X\}$ as in Definition 0.1. Let $T_i \subset X_{i,red}$ be the inverse image of T. We have $X_i \times_X X_{/T} = (X_i)_{/T_i}$ (Lemma 0GVT). Hence $\{(X_i)_{/T_i} \to X_{/T}\}$ is a covering as in Definition 0.1. Moreover, if $X_{red} \setminus T \to X_{red}$ is quasi-compact, so is $X_{i,red} \setminus T_i \to X_{i,red}$ and if X is locally countably indexed, or locally adic*, pr locally Noetherian, the is X_i is countably index, or adic*, or Noetherian. Thus the lemma follows from the affine case which is Lemma 0GVV. □